

Standard Form & Indices

Answer sheet · 41 marks total

Section A — Laws of indices

A1. 4^8 (1)

A2. 7^7 (1)

A3. 6^6 (1)

A4. $2^3 \times 2^2 = 2^5$ (1) = **32** (1)

Section B — Zero and negative indices

B1. 1 (1)

B2. $1 \div 5^2 = 1 \div 25$ (1) = **1/25** (1)

B3. $1 \div 10^3 = 1 \div 1000$ (1) = **0.001** (1)

Section C — Writing standard form

C1. 7.2 (1) = **7.2×10^4** (1)

C2. 8.3 (1) = **8.3×10^6** (1)

C3. 4.8 (1) = **4.8×10^{-3}** (1)

C4. 3 (1) = **3×10^{-5}** (1)

Section D — Converting back, and calculating

D1. **65,000** (1)

D2. **0.00003** (1)

D3. $4 \times 2 = 8$, $10^3 \times 10^5 = 10^8$ (1) = 8×10^8 (1) already in standard form: **8×10^8** (1)

D4. $9 \div 3 = 3$, $10^8 \div 10^3 = 10^5$ (1) = 3×10^5 (1) already in standard form: **3×10^5** (1)

Section E — Powers to know, and estimating roots

E1. **81** (1)

E2. $7^2=49$, $8^2=64$ = between **7 and 8** (1)

Section F — Laws of indices with algebra

F1. **a^{10}** (1)

F2. **b^7** (1)

F3. **c^{12}** (1)

F4. $5 \times 2 = 10$ (1) $d^{2+6} = d^8$ (1) = **$10d^8$**

F5. $3^3 = 27$ (1) $e^{2 \times 3} = e^6$ (1) = **$27e^6$**

Section G — Fractional indices (Higher)

G1. $\sqrt{49} = \mathbf{7}$ (1)

G2. cube root of 27 = $\mathbf{3}$ (1)

G3. 4th root of 16 = 2 (1) $2^3 = \mathbf{8}$ (1)

G4. $\sqrt{9} = 3$ (1) $1 \div 3 = \mathbf{1/3}$ (1)

Marking note: award method marks for a correctly identified front number/power, or correctly applied law of indices, even if the final answer contains an arithmetic error. Accept equivalent fractions/decimals.